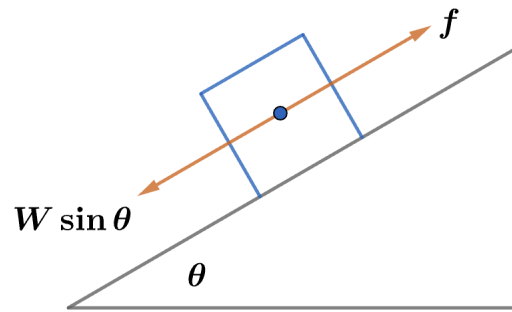


2013 F=ma Exam: Problem 25

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Since the block slides down at constant speed with gravity and friction alone, we have

$$W \sin \theta = f$$

Because $N = W \cos \theta$ and $f = \mu N$,

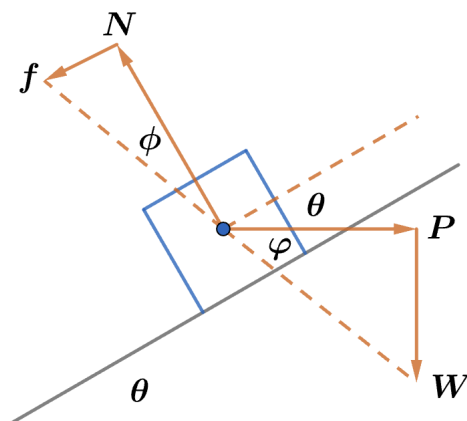
$$W \sin \theta = \mu W \cos \theta$$

Thus,

$$\mu = \tan \theta$$

When a horizontal force P is applied to the box, it moves up the ramp at constant speed so the net force on the box is still zero. Balancing forces,

$$\vec{N} + \vec{f} = \vec{P} + \vec{W}$$



From the free-body diagram, note that $\tan \varphi = W/P$ so

$$P = W \cot \varphi$$

It remains to find φ using the other angles. We have

$$\phi + \theta + \varphi = 90^\circ$$

We are given $\theta = 30^\circ$ and we have

$$\tan \phi = \frac{f}{N} = \mu = \tan \theta$$

which means $\phi = \theta = 30^\circ$ also. Thus,

$$\varphi = 90^\circ - \phi - \theta = 90^\circ - 30^\circ - 30^\circ = 30^\circ$$

$$P = W \cot(30^\circ) = \sqrt{3}W$$

so the answer is D.